Panagiotis Kaloyannis

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Work Experience

Title	Organization/Company	Description	Start—End
Intern ➞ Plasma Control Engineer II	Commonwealth Fusion Systems	Currently developing the SPARC control software and operational procedures. Control work involves designing, implementing, and testing novel real-time capable plasma control and simulation algorithms. Procedural work encompasses the design of the CONOPS for the SPARC control room and plasma experiment proposals. Also significantly contributing to deployment, data acquisition and visualization, machine learning, alarm systems, and facility wide plant system coordination.	2022-Present
Intern	Ubisoft Montreal	Worked to develop fluid physics simulations with hardware independent adaptive resolution for the next generation of video games. Involved both analytical and deep learning methods.	2021-2021
Research Assistant	McGill University	Research assistant at the McGill sTGC lab with Dr. Brigitte Vachon under an NSERC USRA and an IPP CERN Summer Student Fellowship (see Awards and Distinctions). Developed Wire-Support Position Analysis. Also supported Dr. Oscar Hernández and developed CMBStraightStrings. Work was done in C++, ROOT, and Python.	2019-2021

Education & Training

Title	Organization/Company	Description	Start—End
Visiting Student	Massachusetts Institute of Technology (MIT)	Undertook a 6 month research collaboration at the MIT Plasma Science and Fusion Center (PSFC) with Dr. Cristina Rea. Explored the disruptivity metric as a reliable and human interpretable indicator of plasma stability. Thesis analyzed disruption data from the DIII-D and TCV tokamaks. Work largely involved Bayesian statistics and Python optimization. Find the repository here.	2023-2023
MSc. In Applied Physics	École polytechnique fédérale de Lausanne (EPFL)	Graduated as a student in Applied Physics with a Masters Fellowship and a final GPA of 5.41/6. Conducted a collimator geometric optimization study, upgraded the high-speed camera, and did initial scoping of a planned avalanche photodiode upgrade on the Fast Ion Loss Detector (FILD) of the Tokamak à configuration variable (TCV) (now in service). Completed a 12 month internship at Commonwealth Fusion Systems (CFS) as part of the program. Master's thesis was supervised by Dr. Ambrogio Fasoli of EPFL and Dr. Cristina Rea of MIT on the disruptivity metric (repo linked above)	2021-2023
BSc. In Honors Math and Physics	McGill University	Graduated as a student in Honors Mathematics and Physics at McGill University with Dean's Honour List & Distinction. Final GPA of 3.96/4 and collaborated with professors in cosmology and particle physics research. Did extracurricular work in machining and passed machine shop certification.	2018-2021

Talks & Publications

Type	Conference or Jouranl	Title	Year
Talk	APS Division of Plasma Physics (DPP24)	Real time plasma facing component modeling and responses in the SPARC plasma control system.	2024
Poster	APS Division of Plasma Physics (DPP23)	A Deep Dive Into Disruptivity: Learning to Predict and Avoid Disruptions.	2023

Awards & Distinctions

Title	Organization (Location)	Description	Years)
Masters Fellowship	École polytechnique fédérale de Lausanne (EPFL)	Received an EPFL Masters Fellowship, for a total of 40 000 CHF over two years. This grant is given to outstanding applicants to the Faculty of Basic Sciences (FSB) and was renewed upon demonstrated excellence.	2021 2022
Undergraduate Student Research Award (USRA)	NSERC and FRQNT	Granted a USRA by NSERC for work characterizing the LHC at CERN with Professor Brigitte Vachon. The grant paid CA$4500 from NSERC and CA$1500 from the FRQNT.	2020
IPP CERN Summer Student Fellowship	Institute of Particle Physics	Earned a Fellowship to work on upgrades to the LHC at CERN. The scoped project originally included working directly on the ATLAS detector installation but had to be scoped down due to COVID-19.	2020

Extracurriculars

Type	Organization/Company	Description	Start—End
Hackathons & Game Jams	Various Organizations	Competed in many hackathons over the years (mostly with McGill and CERN) where I built things from galactic orbit simulators to snowflake generators (the latter won second place)! Now dipping my toes into game jams and we will see where that takes us!	2017-Present
Sports	Various Leagues	Have competed in both solo and team competitive sports (football, rugby, baseball, hockey, soccer, and ultimate frisbee). Team captain in wrestling, football, and hockey. MVP in Rugby and Wrestling.	2003-Present
Team Robotics	Various Teams	While with the EPFL Xplore mars rover team, built the Astra rover science bay control system. Participated in McGill robotics, was captain of Marianopolis Robotics and Selwyn Robotics. As captain, brought two 2nd place victories home! I may start mentoring in the future so this door is not quite closed yet!	2013-2022

Community Service

Title	Organization/Company	Description	Start—End
Volunteer	Hellenic Appeal Foundation	The Hellenic Appeal Foundation is a charity run by close family friends that does holiday food drives for Easter and Christmas. I helped pack and deliver baskets consistently when I was younger. Now I live in another city and so I help with deliveries if I happen to be around on the special weekend!	2012-Present

Skills and Hobbies

Title	Description
Handywork	I like to work with my hands! Machining, 3D printing, wiring, assembling, woodworking, plumbing, etc.
Baking/Cooking	Cooking is like breathing to me. It is a passion I inherited from my mom. See my recipes!
Programming	I code for a living and enjoy my work. Python, C++, React, Typescript, Rust, Go, CUDA, etc.
Dance	Recently I've taken to the dance floor. Ask me about salsa! I want to learn swing and bachata too!
Languages	I am fluent in French and English. I speak and read Greek. I am learning Spanish!

Contact peter.kaloyannis@outlook.com for sources.